# Kernix 0.7.2 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

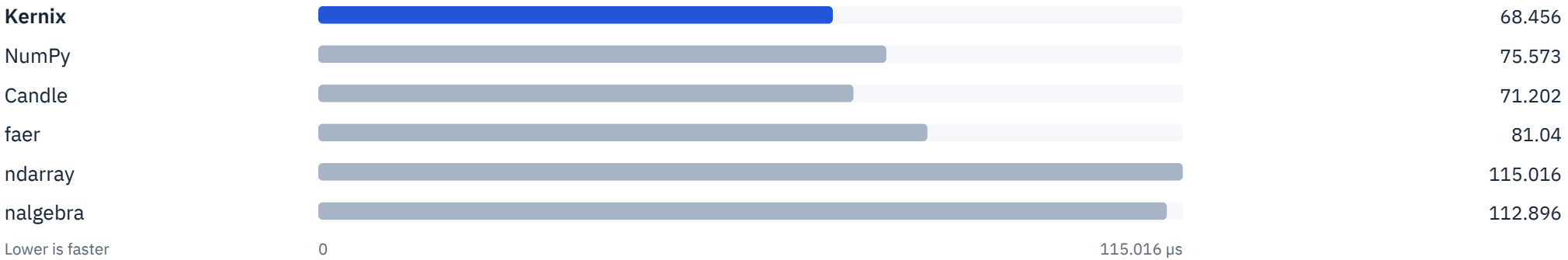
**16** workloads

**62** observations

Source precision retained

## Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




## How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

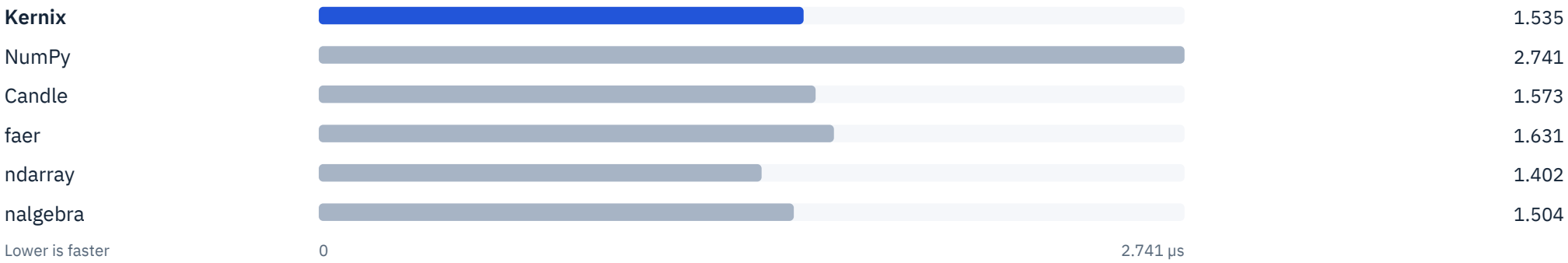
# Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

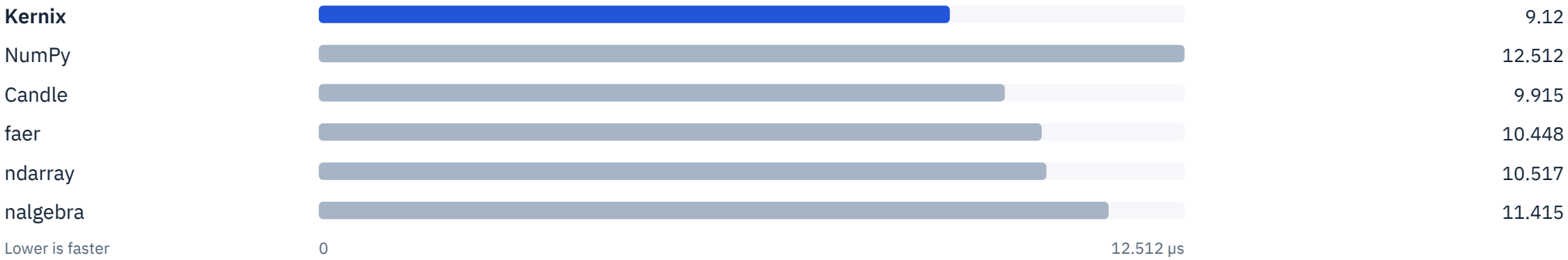
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



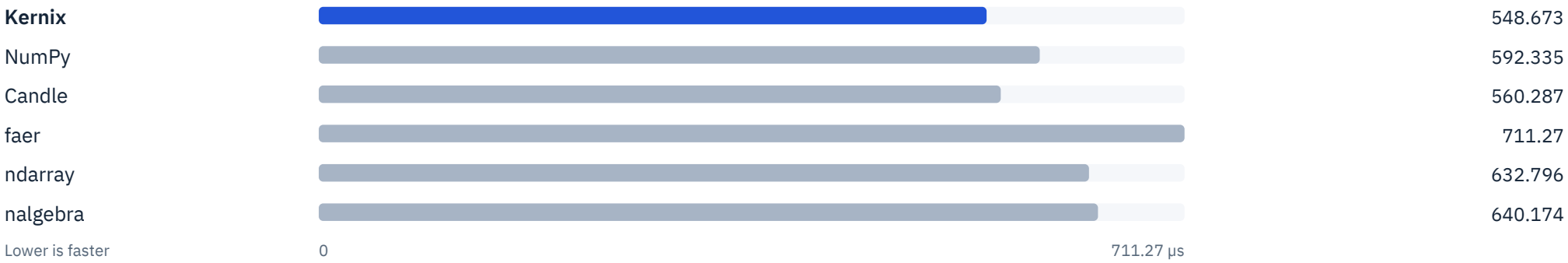
# Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

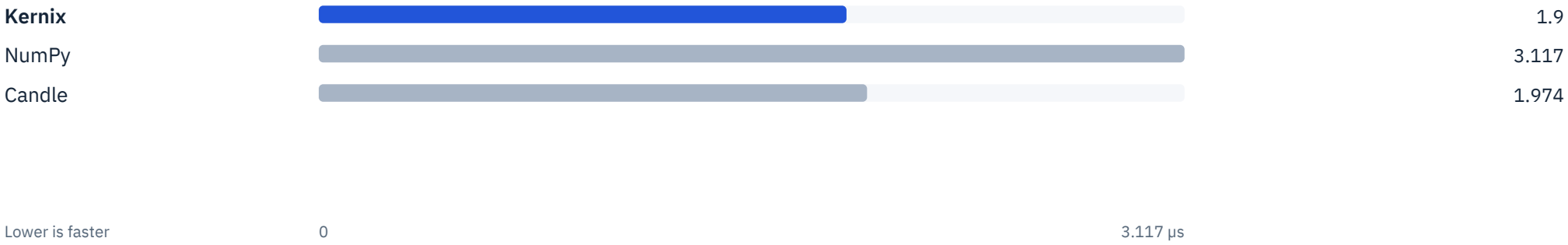
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product




## Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

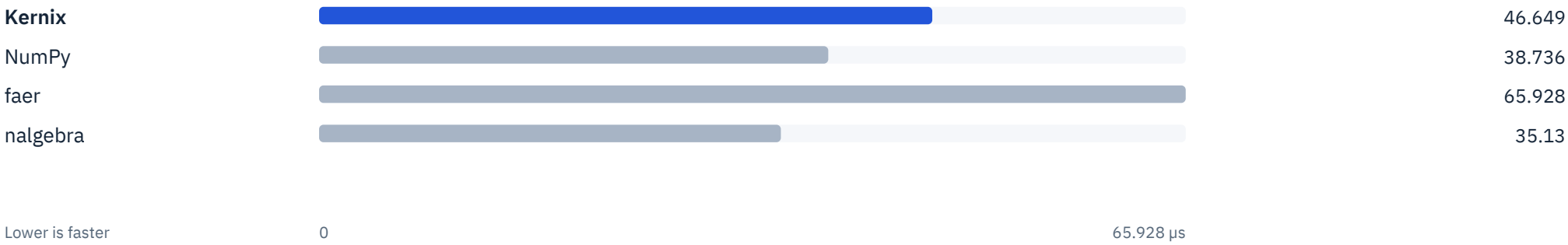


# Kernix 0.7.2 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Thin QR

96 × 32 · f64 · 1 worker · allocated output




## Thin QR

256 × 64 · f64 · 1 worker · allocated output

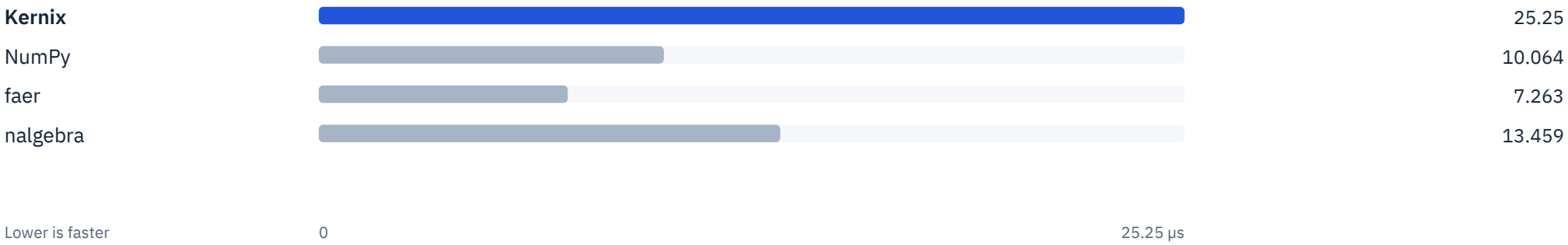


# Kernix 0.7.2 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



# Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product




## Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



# Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model recording + forward

Per-call median

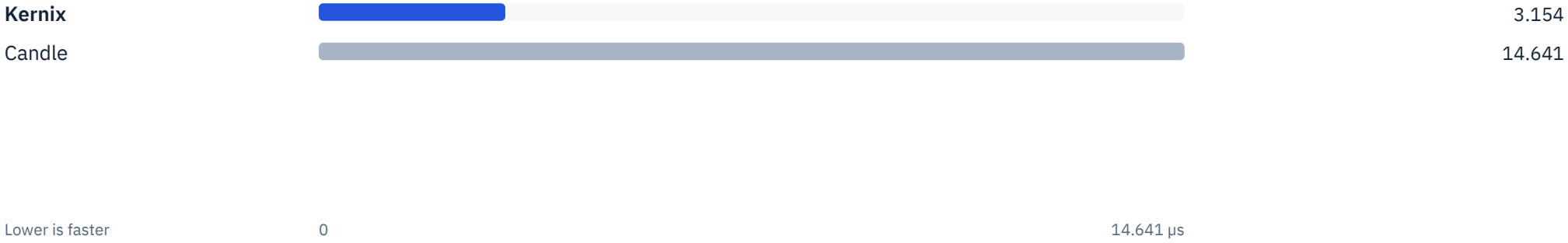
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented




## Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



# Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

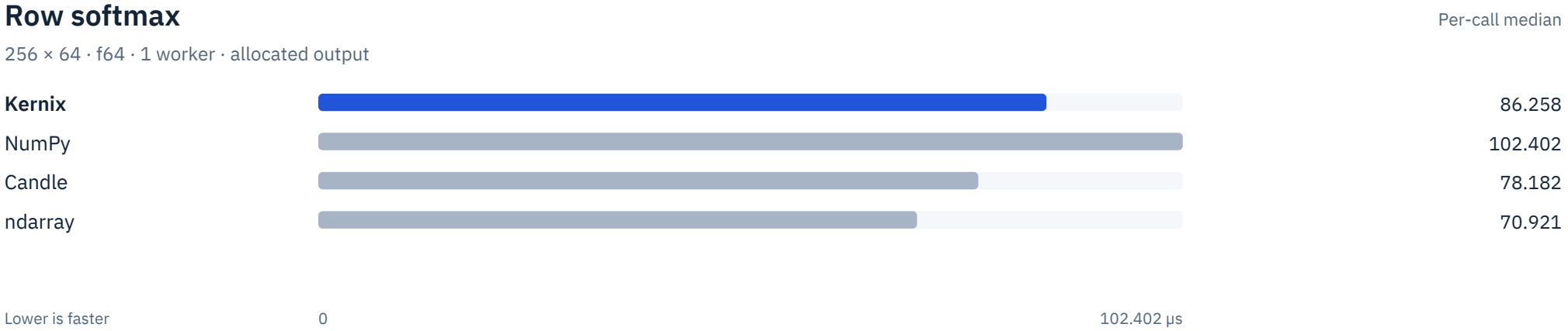


# Kernix 0.7.2 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Row softmax

256 × 64 · f64 · 1 worker · allocated output



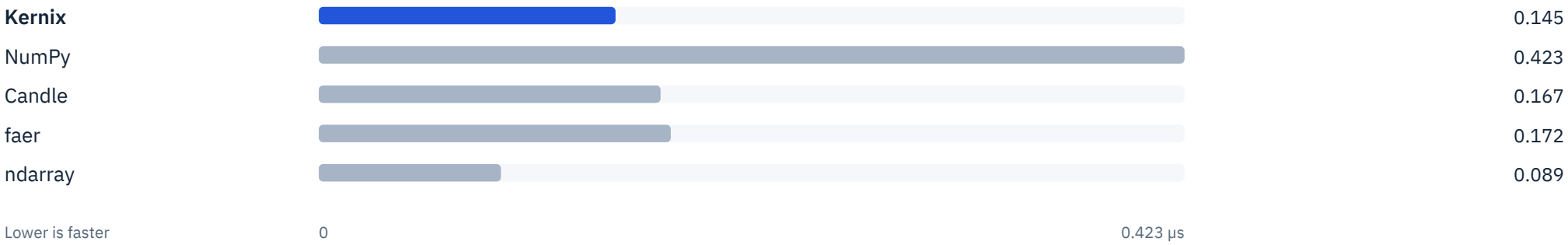
# Kernix 0.7.2 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median




## Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 1535 | 1.535 |
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 1402 | 1.402 |
| **Matrix multiplication / nalgebra**<br>32 × 32 · 32 × 32 · f64 | 1504 | 1.504 |
| **Matrix multiplication / faer**<br>32 × 32 · 32 × 32 · f64 | 1631 | 1.631 |
| **Matrix multiplication / Candle**<br>32 × 32 · 32 × 32 · f64 | 1573 | 1.573 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 2741 | 2.741 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f64 | 9120 | 9.12 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f64 | 10517 | 10.517 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f64 | 11415 | 11.415 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f64 | 10448 | 10.448 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f64 | 9915 | 9.915 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f64 | 12512 | 12.512 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 68456 | 68.456 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 115016 | 115.016 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f64 | 112896 | 112.896 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f64 | 81040 | 81.04 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f64 | 71202 | 71.202 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 75573 | 75.573 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f64 | 548673 | 548.673 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f64 | 632796 | 632.796 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f64 | 640174 | 640.174 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f64 | 711270 | 711.27 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f64 | 560287 | 560.287 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f64 | 592335 | 592.335 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1900 | 1.9 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1974 | 1.974 |
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 3117 | 3.117 |
| **Thin QR / Kernix**<br>96 × 32 · f64 | 46649 | 46.649 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 35130 | 35.13 |
| **Thin QR / faer**<br>96 × 32 · f64 | 65928 | 65.928 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Thin QR / NumPy**<br>96 × 32 · f64 | 38736 | 38.736 |
| **Thin QR / Kernix**<br>256 × 64 · f64 | 487300 | 487.3 |
| **Thin QR / nalgebra**<br>256 × 64 · f64 | 363922 | 363.922 |
| **Thin QR / faer**<br>256 × 64 · f64 | 405668 | 405.668 |
| **Thin QR / NumPy**<br>256 × 64 · f64 | 341148 | 341.148 |
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 25250 | 25.25 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 13459 | 13.459 |
| **Cholesky factorization / faer**<br>64 × 64 · f64 | 7263 | 7.263 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 10064 | 10.064 |
| **Record + backward / Kernix**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 2382 | 2.382 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Record + backward / Candle**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 6678 | 6.678 |
| **Model inference / Kernix**<br>batched normalized/sliced MLP · f64 | 1966 | 1.966 |
| **Model inference / Candle**<br>batched normalized/sliced MLP · f64 | 3479 | 3.479 |
| **Model recording + forward / Kernix**<br>batched normalized/sliced MLP · f64 | 2438 | 2.438 |
| **Model recording + forward / Candle**<br>batched normalized/sliced MLP · f64 | 5299 | 5.299 |
| **Model backward / Kernix**<br>prepared graph → fresh gradients · f64 | 3154 | 3.154 |
| **Model backward / Candle**<br>prepared graph → fresh gradients · f64 | 14641 | 14.641 |
| **Parameter update / Kernix**<br>transactional SGD, fixed gradients · f64 | 489 | 0.489 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 86258 | 86.258 |
| **Row softmax / ndarray**<br>256 × 64 · f64 | 70921 | 70.921 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Row softmax / Candle**<br>256 × 64 · f64 | 78182 | 78.182 |
| **Row softmax / NumPy**<br>256 × 64 · f64 | 102402 | 102.402 |
| **Create zeros / Kernix**<br>64 × 64 · f32 | 145 | 0.145 |
| **Create zeros / ndarray**<br>64 × 64 · f32 | 89 | 0.089 |
| **Create zeros / faer**<br>64 × 64 · f32 | 172 | 0.172 |
| **Create zeros / Candle**<br>64 × 64 · f32 | 167 | 0.167 |
| **Create zeros / NumPy**<br>64 × 64 · f32 | 423 | 0.423 |
| **Create ones / Kernix**<br>64 × 64 · f32 | 185 | 0.185 |
| **Create ones / ndarray**<br>64 × 64 · f32 | 144 | 0.144 |
| **Create ones / faer**<br>64 × 64 · f32 | 201 | 0.201 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Create ones / Candle**<br>64 × 64 · f32 | 237 | 0.237 |
| **Create ones / NumPy**<br>64 × 64 · f32 | 1234 | 1.234 |